

## Project Planning Phase

### Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

|               |                                  |
|---------------|----------------------------------|
| Date          | 03 February 2026                 |
| Team ID       | LTVIP2026TMIDS24253              |
| Project Name  | Electricity-Consumption-Analysis |
| Maximum Marks | 8 Marks                          |

#### PRODUCT BACKLOG, SPRINT SCHEDULE, AND ESTIMATION

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                          | Story Points | Priority | Team Members                 |
|----------|-------------------------------|-------------------|--------------------------------------------------------------------------------------------|--------------|----------|------------------------------|
| Sprint-1 | Data Collection               | USN-1             | As a user, I can load electricity consumption data into the processing environment         | 3            | High     | All                          |
| Sprint-1 | Data Validation               | USN-2             | As a user, I can validate electricity consumption data for accuracy and consistency        | 2            | High     | All                          |
| Sprint-2 | Data Preprocessing            | USN-3             | As a user, I can handle missing values in the electricity consumption dataset              | 3            | Medium   | Team Member 1                |
| Sprint-2 | Data Transformation           | USN-4             | As a user, I can transform electricity consumption data by state and region                | 2            | Medium   | All                          |
| Sprint-3 | Tableau Visualizations        | USN-5             | As a user, I can create 12 professional Tableau visualizations for electricity consumption | 8            | High     | Team Member 1, Team Member 2 |
| Sprint-3 | Interactive Dashboard         | USN-6             | As a user, I can build interactive dashboard with filtering capabilities                   | 5            | High     | Team Member 2, Team Member 3 |
| Sprint-4 | Web Application               | USN-7             | As a user, I can develop Flask web application with                                        | 6            | High     | All                          |

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                           | Story Points | Priority | Team Members                 |
|----------|-------------------------------|-------------------|-----------------------------------------------------------------------------|--------------|----------|------------------------------|
|          |                               |                   | responsive design                                                           |              |          |                              |
| Sprint-4 | Analysis Scenarios            | USN-8             | As a user, I can implement three analysis scenarios with focused insights   | 4            | Medium   | Team Member 3, Team Member 4 |
| Sprint-5 | Documentation                 | USN-9             | As a user, I can create comprehensive project documentation and user guides | 5            | Medium   | All                          |
| Sprint-5 | Testing & Deployment          | USN-10            | As a user, I can perform testing and deploy project to GitHub repository    | 3            | High     | All                          |

#### PROJECT TRACKER, VELOCITY & BURNDOWN CHART

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|-------------------------------------------------|------------------------------|
| Sprint-1 | 5                  | 1 Week   | 01 February 2026  | 07 February 2026          | 5                                               | 07 February 2026             |
| Sprint-2 | 5                  | 1 Week   | 08 February 2026  | 14 February 2026          | 5                                               | 14 February 2026             |
| Sprint-3 | 13                 | 1 Week   | 15 February 2026  | 21 February 2026          | 13                                              | 21 February 2026             |
| Sprint-4 | 10                 | 1 Week   | 22 February 2026  | 28 February 2026          | 10                                              | 28 February 2026             |
| Sprint-5 | 8                  | 1 Week   | 01 March 2026     | 07 March 2026             | 8                                               | 07 March 2026                |

---

## VELOCITY CALCULATION

### Team Velocity Analysis:

- Total Project Duration: 5 weeks (35 days)
- Total Story Points: 41 points
- Average Velocity per Sprint: 8.2 points per sprint
- Average Velocity per Day: 1.17 points per day

### Velocity Formula:

Average Velocity (AV) = Total Story Points ÷ Total Duration

AV = 41 points ÷ 35 days = 1.17 points per day

### Team Capacity:

- Team Size: 4 members
- Sprint Duration: 1 week per sprint
- Expected Velocity: 8-10 points per sprint
- Actual Performance: Consistent completion of all planned story points

---

## BURNDOWN CHART

**Burndown Chart Description:** A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

### Project Burndown Analysis:

- Initial Work: 41 story points
- Sprint 1: 36 points remaining (5 completed)
- Sprint 2: 31 points remaining (5 completed)
- Sprint 3: 18 points remaining (13 completed)
- Sprint 4: 8 points remaining (10 completed)
- Sprint 5: 0 points remaining (8 completed)

### Burndown Pattern:

- **Steady Progress:** Consistent completion across all sprints
  - **Peak Performance:** Sprint 3 with highest completion (13 points)
  - **Efficient Delivery:** All sprints completed on time
  - **Zero Backlog:** All story points completed by project end
- 

## **SPRINT DETAILS**

### **Sprint 1: Foundation (5 points)**

- **Focus:** Data collection and validation
- **Duration:** 1 week
- **Team:** All members
- **Outcome:** Clean, validated dataset ready for processing

### **Sprint 2: Data Processing (5 points)**

- **Focus:** Data preprocessing and transformation
- **Duration:** 1 week
- **Team:** Distributed workload
- **Outcome:** Structured data ready for visualization

### **Sprint 3: Visualization Development (13 points)**

- **Focus:** Tableau visualizations and dashboard
- **Duration:** 1 week
- **Team:** Core visualization team
- **Outcome:** 12 professional visualizations and interactive dashboard

### **Sprint 4: Web Application (10 points)**

- **Focus:** Flask application and analysis scenarios
- **Duration:** 1 week
- **Team:** All members with specialized roles
- **Outcome:** Complete web application with responsive design

### **Sprint 5: Finalization (8 points)**

- **Focus:** Documentation, testing, and deployment
  - **Duration:** 1 week
  - **Team:** All members
  - **Outcome:** Professional project ready for delivery
- 

### **TEAM PERFORMANCE METRICS**

#### **Velocity Metrics:**

- **Consistent Velocity:** 8.2 points average per sprint
- **100% Completion Rate:** All sprints completed successfully
- **On-Time Delivery:** All releases on planned dates
- **Quality Assurance:** All acceptance criteria met

#### **Efficiency Indicators:**

- **No Delays:** Zero sprint delays
  - **Full Story Point Completion:** 100% completion rate
  - **Team Collaboration:** Effective distributed workload
  - **Risk Management:** No blockers or impediments
- 

### **PROJECT SUCCESS FACTORS**

#### **Agile Practices:**

- **Sprint Planning:** Effective story point estimation
- **Daily Standups:** Regular progress tracking
- **Sprint Reviews:** Continuous stakeholder feedback
- **Retrospectives:** Process improvement opportunities

#### **Quality Assurance:**

- **Definition of Done:** Clear acceptance criteria

- **Testing:** Comprehensive validation at each sprint
  - **Documentation:** Continuous documentation updates
  - **Deployment:** Regular integration and deployment
- 

## **RESOURCE ALLOCATION**

### **Team Member Distribution:**

- **Team Member 1:** Data processing and Tableau visualizations
- **Team Member 2:** Dashboard development and web integration
- **Team Member 3:** Flask application and analysis scenarios
- **Team Member 4:** Documentation, testing, and deployment

### **Skill Utilization:**

- **Data Analysis:** 40% of team effort
  - **Visualization:** 30% of team effort
  - **Web Development:** 20% of team effort
  - **Documentation:** 10% of team effort
- 

## **RISK MITIGATION**

### **Identified Risks:**

- **Data Quality Issues:** Mitigated through validation processes
- **Technical Complexity:** Managed through distributed expertise
- **Timeline Constraints:** Addressed through efficient sprint planning
- **Integration Challenges:** Resolved through iterative development

### **Mitigation Strategies:**

- **Buffer Time:** Built into sprint planning
- **Cross-training:** Team members skilled in multiple areas
- **Regular Monitoring:** Daily progress tracking

- **Contingency Planning: Alternative approaches prepared**
- 

## **PROJECT DELIVERY**

### **Final Deliverables:**

- **Complete Dataset: Processed and validated electricity consumption data**
- **Professional Visualizations: 12 interactive Tableau charts**
- **Web Application: Flask application with responsive design**
- **Comprehensive Documentation: User guides and technical documentation**
- **GitHub Repository: Professional project presentation**

### **Success Criteria:**

- **100% Story Point Completion: All requirements fulfilled**
- **On-Time Delivery: All sprints completed as planned**
- **Quality Standards: Professional implementation achieved**
- **Stakeholder Satisfaction: All acceptance criteria met**

**Feedback submitted**

**Command Awaiting Approval**

**Drop to add to Cascade**